

Sorting

Ascending (smaller to bigger) {5, 7, 9, 11, 13}

Descending (bigger to smaller) {13, 11, 9, 7, 5}

It is the process of arranging data or items in a specific order usually ascending or descending.

Example → {"banana", "apple", "cherry"}

↓

{"apple", "banana", "cherry"}

Example → {3, 1, 4, 2, 7, 9, 8}

Ascending



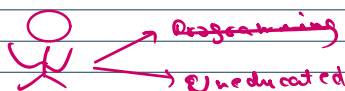
$arr[i] \leq arr[i+1]$

↓
set i to right
↓
bigger element

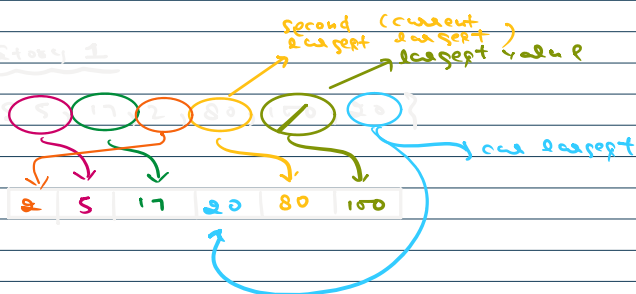
Stories to remember



Coding blocks



Story 1



1. Pick highest marks → Place them at last position. {5, 17}
2. Second highest marks → Place them at second last position. {2, 80} → largest
3. I will get Rank of all the students

Bubble Sort

"Bubble the maximum at the last"

$arr[i] > arr[i+1]$
↓
swap

arr → 1 5 2 0 4

0 1 2 3 4

{+len-1}



arr → 1 5 2 6 4

loop

{ + len = 1 }

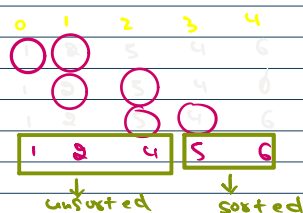


$arr[0] > arr[1] \Rightarrow 1 > 5 \Rightarrow \times$
 $arr[1] > arr[2] \Rightarrow 5 > 2 \Rightarrow \checkmark \Rightarrow \text{swap}$
 $arr[2] > arr[3] \Rightarrow 5 > 6 \Rightarrow \times$
 $arr[3] > arr[4] \Rightarrow 6 > 4 \Rightarrow \checkmark \Rightarrow \text{swap}$

→ max ele bubble out at last

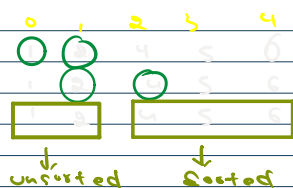
$arr[2] > arr[3]$
 $\downarrow$
 swap

{ + len = 2 }



$arr[0] > arr[1] \Rightarrow 1 > 2 \Rightarrow \times$
 $arr[1] > arr[2] \Rightarrow 2 > 5 \Rightarrow \times$
 $arr[2] > arr[3] \Rightarrow 5 > 4 \Rightarrow \checkmark \rightarrow \text{swap}$

{ + len = 3 }



$arr[0] > arr[1] \Rightarrow 1 > 2 \Rightarrow \times$
 $arr[1] > arr[2] \Rightarrow 2 > 4 \Rightarrow \times$

{ + len = 4 }



$arr[0] > arr[1] \Rightarrow 1 > 2 \Rightarrow \times$

{ + len = 1 }

n = 5

loop → 0 to 3

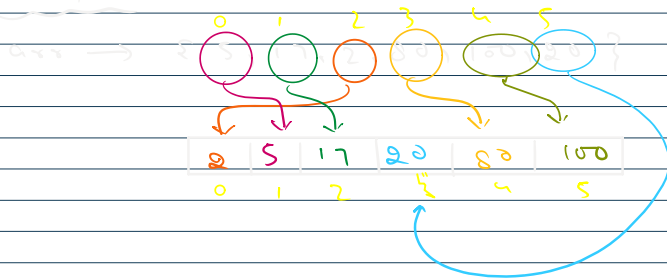
$arr[0] > arr[1] \Rightarrow 1 > 5 \Rightarrow \times$
 $arr[1] > arr[2] \Rightarrow 5 > 2 \Rightarrow \checkmark \Rightarrow \text{swap}$
 $arr[2] > arr[3] \Rightarrow 5 > 6 \Rightarrow \times$
 $arr[3] > arr[4] \Rightarrow 6 > 4 \Rightarrow \checkmark \Rightarrow \text{swap}$

$\downarrow$
 loop → 0 to n - len

for (int i = 0; i < n - len; i++) {
 if (arr[i] > arr[i + 1]) {
 // swap arr[i], arr[i + 1]
 }
 }



Story 2

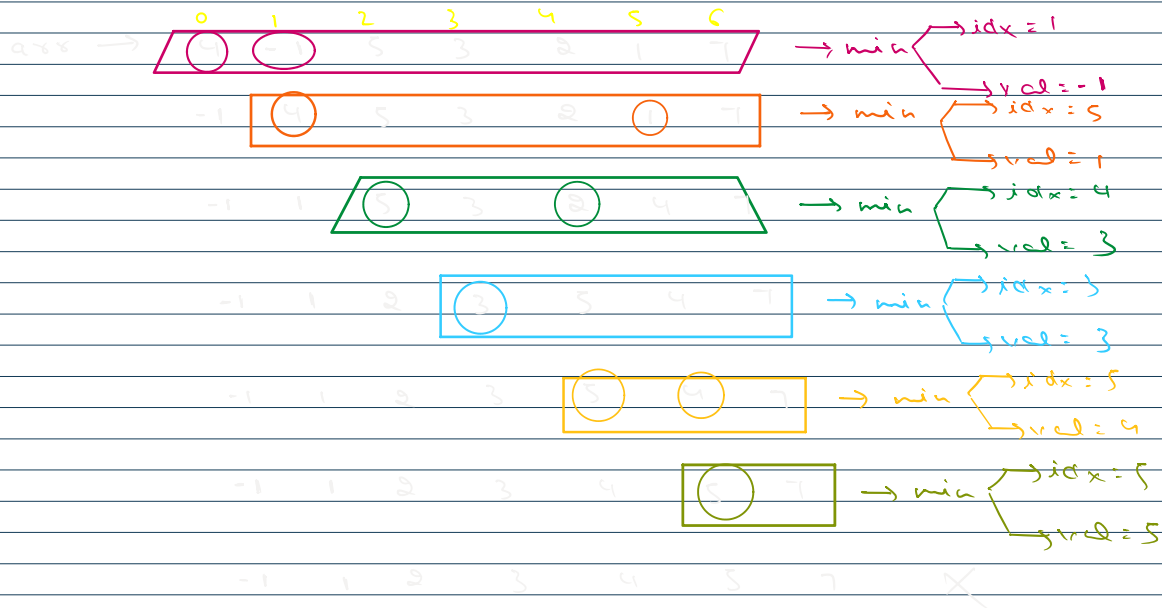


1. Pick the lowest marks $\Rightarrow$ Place at first position $\{0\}$
2. Pick the second lowest marks $\Rightarrow$ Place at second position $\{1\}$ $\Rightarrow$ lowest
3. We will get rank of all the students.

Can you find the index minimum value in an array of length n , starting from a given index

Selection Sort

"Put minimum element at first position"



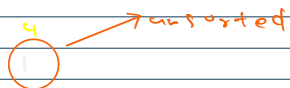
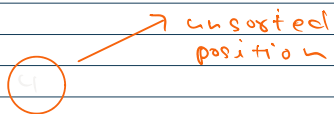
1. Go to every position
2. Find min index $\Rightarrow$ swap both



Story 3

school assembly.

Insertion Sort



can you move the last element in the array to its correct position?

Answer 1)



Can you move the next element in the array to its sorted position?

Step 1

	0	1	2	3	4
$j=3$	2	3	4	5	5
	2	3	4	1	5

$5 > 1 \Rightarrow \checkmark$
 $arr[4] = 5$
 $arr[3] = 1$

$arr[j] > item$
 $\downarrow$
 $arr[j+1] = arr[j]$
 $arr[j] = item$

	0	1	2	3	4
$j=2$	2	3	4	4	5
	2	3	1	4	5

$4 > 1 \Rightarrow \checkmark$
 $arr[3] = 4$
 $arr[2] = 1$

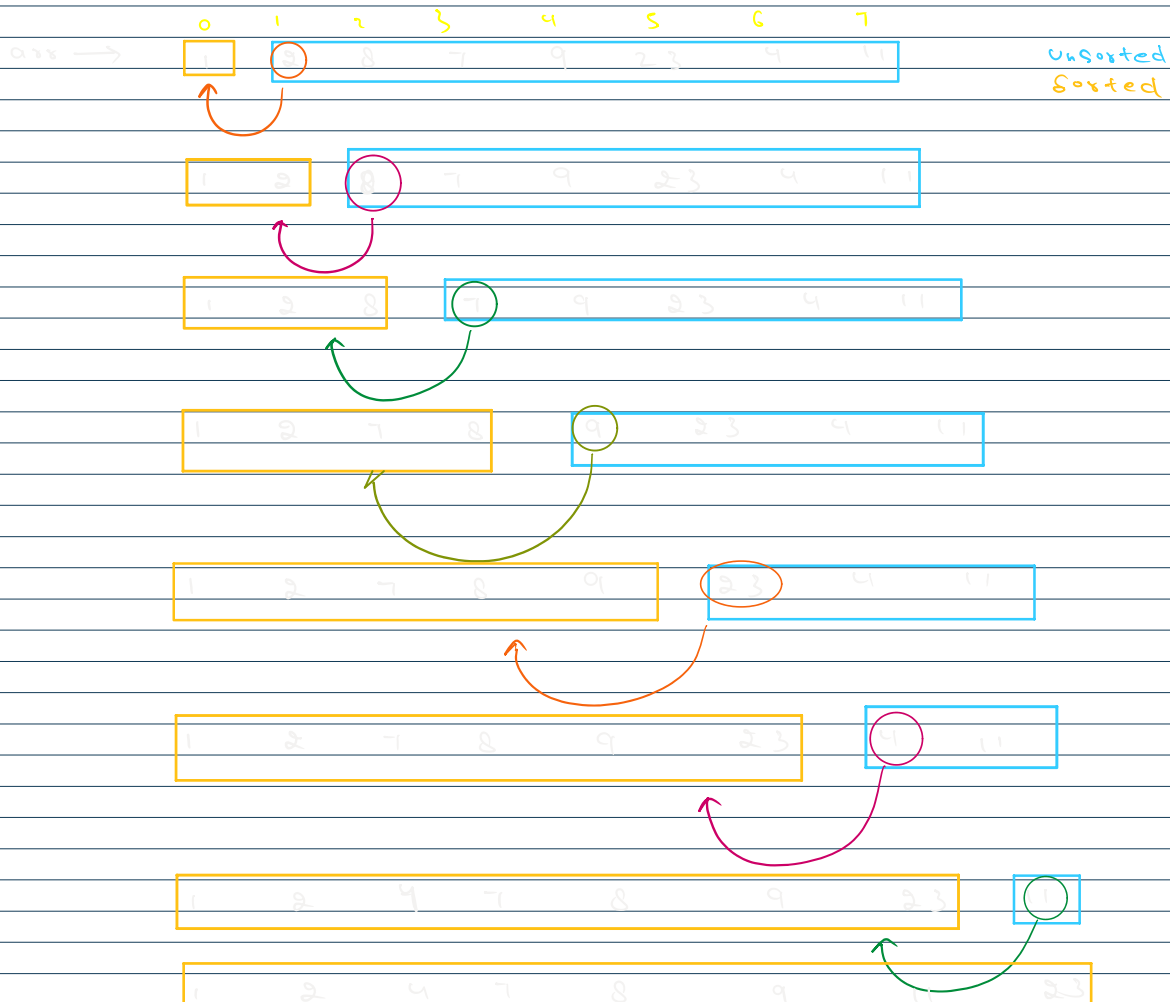
	0	1	2	3	4
$j=1$	2	3	3	4	5
	2	1	3	4	5

$3 > 1 \Rightarrow \checkmark$
 $arr[2] = 3$
 $arr[1] = 1$

	0	1	2	3	4
$j=0$	2	2	3	4	5
	1	2	3	4	5

$2 > 1 \Rightarrow \checkmark$
 $arr[1] = 2$
 $arr[0] = 1$

sorted position





Subarray

contiguous part of an array

arr $\rightarrow$ [7, 1, 2]

subarrays $\rightarrow$ []

7
1
7 1
1 2
7 1 2
~~7 2~~ x
~~7 1 2~~ x
~~1 2~~ x

Maximum Sub-Array

Panna
malala
(+5)

Idli
(-3)

Q. Lab
Tamen
(+2)

No. Spill
dal
(-8)

Plain
rice
(0)

Butter
Chicken
(+7)

Ice cream
(+9)



+5 - 3 + 2 - 8 + 0 + 7 + 9 = 8

score := 24 - 16 = 8

max score

max sum of subarray

sin water
(-1)

me